

JIAYI WU

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Google Scholar ◊ GitHub ◊ LinkedIn ◊ Personal Webpage

EDUCATION

Brown University

Sept 2024 - May 2028 (expected)

A.B. Mathematics and Sc.B. Computer Science

Advisor: Isabel Vogt and Suresh Venkatasubramanian

Selected Coursework: Advanced Topics in Deep Learning, Graduate Deep Learning, Topics in High-Dimensional Probability, Theory of Computation, Formal Proof and Verification, Using an Interactive Proof Assistant to Do Mathematics, Mathematical Logic, Probabilistic Methods in Computer Science, Critical Data and Machine Learning, Graduate Real Analysis, Abstract Algebra

RESEARCH INTERESTS

I hope to build the *formal* and *empirical* infrastructure for **trustworthy machine learning** across two complementary dimensions:

- **AI for mathematics and automated reasoning:** Formal theorem proving and program synthesis are fully specified, unambiguous, machine-checkable substrates for studying models' reasoning processes. I'm interested in designing model architectures that leverage various learning paradigms (neurosymbolic programming in particular) to enable more robust and interpretable reasoning, both in theorem proving and beyond.
- **Principled evaluation, auditing, and guarantees:** I'm interested in designing evaluation and auditing approaches that attach formal or statistical evidence to model properties, from specification conformance and robustness to value alignment and supply-chain provenance. These approaches should generalize across systems, development stages, and deployment contexts.

PUBLICATIONS

* represents equal contribution.

- [1] **Jiayi Wu**, R. J. George, and A. Anandkumar, "ITPEval: Benchmarking formal translation across interactive theorem provers," in *ICML 2026 3rd AI for Math Workshop: Toward Self-Evolving Scientific Agents*, 2026. [Online]. Available: <https://openreview.net/forum?id=4PShGfBOMG>
- [2] I. Ivanov, **Jiayi Wu**, G. Zhao, Z. Li, S. Bach, and R. Lewis, "Mining machine-generated lean proofs: Advice for users, developers, and testers," in *Conference on Artificial Intelligence and Theorem Proving (AITP 2026)*, 2026.
- [3] E. Poole-Dayana, **Jiayi Wu**, T. Sorensen, J. Pei, and M. A. Bakker, "Benchmarking overton pluralism in LLMs," in *The Fourteenth International Conference on Learning Representations (ICLR 2026)*, also accepted to *NeurIPS 2025 Workshop on Evaluating the Evolving LLM Lifecycle: Benchmarks, Emergent Abilities, and Scaling*, 2025. [Online]. Available: <https://arxiv.org/abs/2512.01351>
- [4] C. J. Li*, **Jiayi Wu***, Z. Mo, A. Qu, Y. Tang, K. I. Zhao, Y. Gan, J. Fan, J. Yu, J. Zhao, P. P. Liang, L. A. A. Pastor, and K. Larson, "Simulating society requires simulating thought," in *The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025) Position Paper Track*, 2025. [Online]. Available: <https://arxiv.org/abs/2506.06958>
- [5] C. J. Li*, Z. Mo*, Y. Tang, A. Qu, **Jiayi Wu**, K. I. Zhao, Y. Gan, J. Fan, J. Yu, H. Jiang, P. P. Liang, J. Zhao, L. A. A. Pastor, and K. Larson, "HugAgent: Evaluating LLMs in simulating individual-level human reasoning on open-ended tasks," in *NeurIPS 2025 Workshop on Bridging Language, Agent, and World Models for Reasoning and Planning (Spotlight)*, also accepted to *NeurIPS 2025 Workshop on PersonaLLM: Workshop on LLM Persona Modeling (Oral)* and *NeurIPS 2025 Workshop on Socially Responsible and Trustworthy Foundation Models (ResponsibleFM)*, 2025. [Online]. Available: <https://arxiv.org/abs/2510.15144>

RESEARCH EXPERIENCE

Interoperability for Formal Mathematics: Automating Proof Translation Across ITPs [1] Feb 2026 - Present
California Institute of Technology Pasadena, CA

Summer Undergraduate Research Fellowships (SURF) project advised by Anima Anandkumar, in collaboration with Robert Joseph George.

Neurosymbolic Inference of Regular Stream Types Jan 2026 - Present
Brown University Providence, RI

Undergraduate Teaching and Research Awards (UTRA) project advised by Nikos Vasilakis, in collaboration with Lukas Lazarek, Zekai Li, and Evangelos Lamprou.

Pattern Mining and Automated Tactic Discovery in Theorem Proving [2] Oct 2025 - Present
Brown University Providence, RI

Research project (CSCI 1973 Individual Independent Study) advised by Robert Lewis and Stephen Bach, in collaboration with Gavin Zhao, Zekai Li, and Ilija Ivanov.

Formalizing and Benchmarking Overton Pluralism in Large Language Models [3] May 2025 - Dec 2025
Massachusetts Institute of Technology Cambridge, MA

Visiting summer researcher project advised by Michiel Bakker, in collaboration with Elinor Poole-Dayana.

Motif of Thoughts (MoT): Reusable Abstractions for Neurosymbolic Reasoning [4] [5] Mar 2025 - Present
Massachusetts Institute of Technology Cambridge, MA

Research project in collaboration with Chance Jiajie Li.

LEADERSHIP & TEACHING

Undergraduate Teaching Assistant, CSCI 2470 Graduate Deep Learning Aug 2026 - Present

Co-Organizer, CNTR Technology & Science Policy (TASP) Summit May 2026 - Present

Student Liaison, Center for Technological Responsibility (CNTR) May 2026 - Present

Undergraduate Teaching Assistant, CSCI 1952B Responsible Computer Science in Practice Jan 2026 - May 2026

Co-President, AI Robotics Ethics Society (AIRES) at Brown Sep 2025 - May 2026

Undergraduate Teaching Assistant, CSCI 0170 Introduction to Computer Science Aug 2025 - Dec 2025

AI Governance Panel Director, Brown China Summit Sep 2024 - Mar 2025

TECHNICAL SKILLS

Programming Languages Java, Python, JavaScript, HTML/CSS, SQL, MATLAB, Lean, Isabelle, ReasonML, Racket, C/C++, Processing

Programming Tools PyTorch, TensorFlow, scikit-learn, NumPy, pandas, Git, Linux